



Albert Einstein Cup 2026

Skill 08: Mobile Applications Development

Session 5: NeubrandenBook Reading & Quotes (Development & Testing)

1. Introduction

NeubrandenBook introduces a lightweight reading module to allow users to:

- track reading progress for a selected book
- extract and view quotes from the book content

In this module, you need to develop the basic App functional logic and write automated tests to run the App.

The wireframes (Wireframes/) are for reference only. You do not need to replicate the exact appearance, colors, decorative icons/images, or elements' positions/styles.

You have **1:30 hours** to implement the app.

2. General Demands

1. Use the data provided in Data/.
2. Your application functions should be developed according to all test cases.
3. User data must not be persisted across app restarts.

3. Develop by Test Case

Develop the application by understanding the test cases.

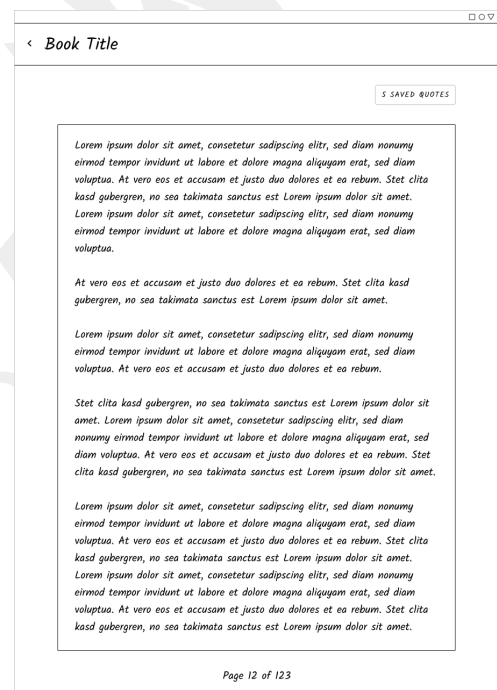
3.1. Book List Page

1. The title area contains the NeubrandenBook logo and the title "My Library".
2. Each book list item includes:
 1. Title
 2. Author
 3. Total Pages
 4. Quote Count, only if the user has saved at least one quote from this book
 5. Current Page Number
 6. Reading Progress Bar
3. Show the Current Page Number and Reading Progress Bar only if the user has started reading the book, i.e., is at least on page two.
4. Click on a book to navigate to the corresponding Reading Page.



3.2. Reading Page

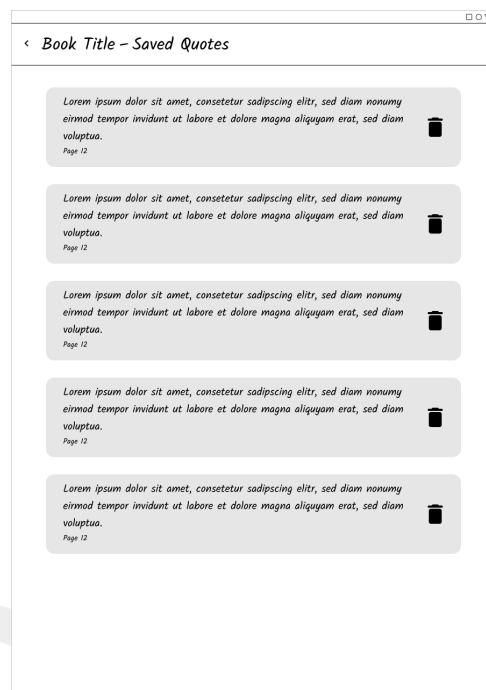
1. The title bar contains a back button and the book title.
2. When returning to the Book List Page, the Current Page Number and Reading Progress Bar for the book are updated according to the page the user left off on.
3. The content area shows:
 1. Saved Quotes Button ("1 Saved Quote" / "X Saved Quotes"), only if the user has saved at least one quote from this book
 2. Book's Pages
 3. Reading Progress ("Page X of Y")
4. Click on the Saved Quotes Button to navigate to the Saved Quotes Page for this book.
5. The user can navigate through the book pages by swiping left or right.
6. The initial page is the page where the user last left off. If the user has not started reading the book, start on page 1.
7. Long press and drag to select text on a page.
 1. When text is selected, a Save Quote Button appears.
 2. Click on the Save Quote Button to save the quote and update the number in the Saved Quotes Button accordingly.



- Text corresponding to saved quotes is highlighted on the page.

3.3. Saved Quotes Page

- The title bar contains a back button and the title "Book Title – Saved Quotes".
- Each quote entry includes:
 - Quote text
 - Page Number where the quote was extracted from
 - Delete Button
- Quotes are sorted by their page number in ascending order.
- Click on a quote to navigate back to the Reading Page, which opens on the corresponding page number.
- Click on the Delete Button to remove the quote from the list and update the number of saved quotes shown on the Reading Page accordingly.



4. Automated Test

Finish the automated test script according to the description below.

The test script must be stored and be runnable according to 5. Automated Test Guide (page 4).

All steps need to be held on for at least 10 seconds, and "Step <No.>: <Description>" should be logged in the console.

No.	Area	Description	Expectation
1		Launch the app.	The app starts normally.
2	Book List Page	Click on the book "Schatz der Welten: Untergang".	The Reading Page for "Schatz der Welten: Untergang" opens, showing page 1.
3	Reading Page	Swipe to go to page 2.	Page 2 is displayed, and the reading progress shows "Page 2 of 391".
4	Reading Page	Long press and drag to select the first real paragraph ("Insbesondere sein [...] stark werden.").	The first real paragraph is highlighted, and a Save Quote Button appears.
5	Reading Page	Click the Save Quote Button	The quote is saved, the Save Quote Button disappears, and the Saved Quotes Button appears ("1 Saved Quote").
6	Reading Page	Swipe to go to page 3.	Page 3 is displayed, and the reading progress shows "Page 3 of 391".
7	Reading Page	Long press and drag to select the third real paragraph ("Sofort kam [...] was eigentlich?").	The third real paragraph is highlighted, and a Save Quote Button appears.

No.	Area	Description	Expectation
8	Reading Page	Click the Save Quote Button.	The quote is saved, the Save Quote Button disappears, and the Saved Quotes Button updates ("2 Saved Quotes").
9	Reading Page	Click the Saved Quotes Button.	The Saved Quotes Page opens, showing the two saved quotes with their corresponding page numbers.
10	Saved Quotes Page	Click the Delete Button for the second quote.	The second quote is removed from the list.
11	Saved Quotes Page	Click on the remaining quote.	The Reading Page opens on the second page with the quote highlighted.
12	Reading Page	Click the back button.	The Book List Page opens. The book "Schatz der Welten: Untergang" shows a Quote Count of 1, the Current Page Number as 2, and the Reading Progress Bar reflects this progress.

5. Automated Test Guide

In the `pubspec.yaml` file of your Flutter app, add the `flutter_test` and `integration_test` packages to your `dev_dependencies`:

```
dev_dependencies:
  flutter_test:
    sdk: flutter
  integration_test:
    sdk: flutter
```

Write your test scripts in the file `integration_test/app_test.dart`. Make sure that the test can be started by running the following console commands:

1. `flutter pub get --offline`
2. `flutter test --no-pub integration_test/app_test.dart`

6. Submission

Your submission should contain the following top-level files and directories:

- the compiled APK with package name `de.blzit.aec2026.skill08.session5.competitorX`
- a directory with the full source code
- optional: a README file containing notes for the marking

These files and directories should be pushed to the Git repository `Session-5_Competitor8-X`. Only the latest contents of the `main` branch will be taken into account.